now the West Bank, Azzam would have suffered through the Palestinian defeat in the first Arab-Israeli war of 1947–48, which Palestinians refer to as the Naqba, or catastrophe. At its conclusion, Israel achieved independence, while Palestine was divided between Gaza, which came under Egypt's control, and the West Bank, which was annexed to the Hashemite Kingdom of Jordan. As a result, hundreds of thousands of Palestinians lost their homes and became refugees.

In 1963 Azzam moved to Syria to attend Damascus University, where in 1966 he obtained a degree in Islamic law. Here he was greatly influenced by faculty belonging to the Muslim Brotherhood, the oldest Islamist party in the Arab world, though he probably was already a member from his youth in Jordan.[35] After graduation he returned to the West Bank, just before the June 1967 Arab-Israeli war. Israel's swift conquest of the West Bank led to another flow of Palestinian refugees fleeing Israeli rule; Azzam's family joined the exodus to Jordan's East Bank.

They settled in a refugee camp outside the city of Zarqa, which would become a hotbed of Islamism over the next half century. Azzam joined the Palestinian fedayeen resistance movement and participated in their attempts to attack Israelis across the Jordan. Though an active fighter in the fedayeen forces, he wanted to continue his studies so moved to Cairo, Egypt, in late 1968, where he obtained a master's degree with honors at the Islamic world's premier institution of higher learning, Al Azhar University. He then took up a teaching position at Jordan University but apparently sat out Jordan's civil war in 1970, not fighting on either the fedayeen or the government side.

In 1971 he returned to Al Azhar for his doctorate, awarded in 1973. In Cairo he became further embedded in the underground world of the Muslim Brotherhood and the Islamist movement. Upon graduation, he went back to teaching at the university in Amman, also becoming well known in Jordan as an Islamic leader and speaker. King Hussein's government tolerated the Muslim Brotherhood and the Islamists in order to keep the more radical nationalist Palestinian movements in check, but it also watched their activists closely to ensure they did not get out of control and pose a threat to the Hashemites.

Azzam's impressive credentials as an Islamic scholar and his increasingly radical views attracted the attention of Jordan's intelligence service, which, like the Saudi service, was called the General Intelligence